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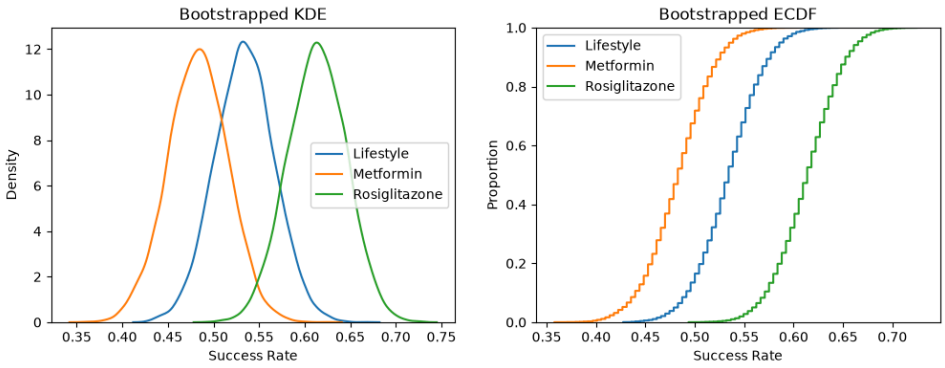
Trial Summary

Background

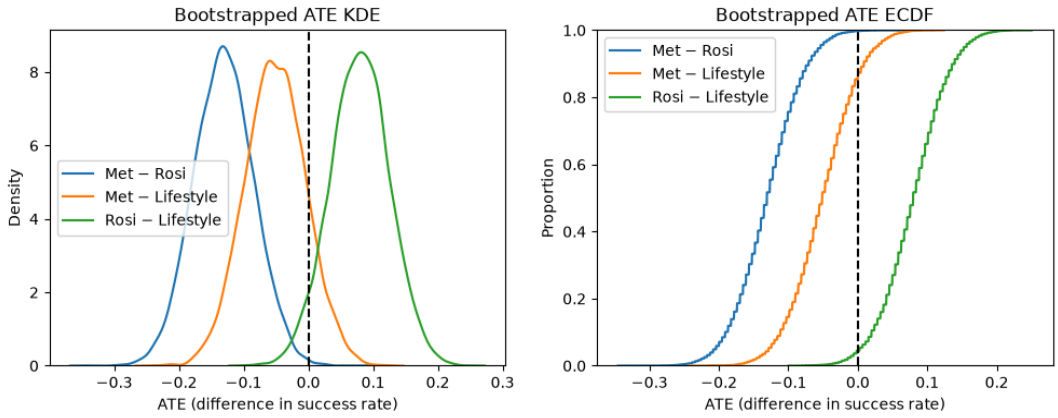
The goal of this trial was to compare how three treatments reduced the full progression of diabetes in pre-diabetics. Pre-diabetic participants were assigned to one of three treatments: metformin, rosiglitazone, and lifestyle adjustments. A patient was labeled a success if, after the trial, the model measured a reduction in the full progression of diabetes. If none was measured, they were labeled a failure. To allow the model to interpret the predictors, I encoded success as 1 and failure as 0. Since success is binary, you are either successful or not; encoding it as 0 and 1 accurately represents the situation.

Design & Results

First, I calculated the mean for each group and found that rosiglitazone was highest at 61%, followed by lifestyle adjustments and metformin at 53% and 48%, respectively. To ensure the variation between groups was not due to random noise, I bootstrapped by resampling within each group with replacement 10,000 times. This produced 10,000 new estimated success rates for each treatment, shown in the KDE and corresponding ECDF below.



Notice that while the distributions have similar spreads, the KDE center for rosiglitazone sits noticeably higher than the other two. To test whether the treatment differences are real and not due to chance, I used a bootstrapping procedure on the ATE (average treatment effect). In every iteration of the 10,000 bootstrap run, the difference between the means of each predictor is computed for each pair of predictors. The KDE and corresponding ECDF plot these 10,000 sets of differences below.



We cannot say there is a meaningful difference between two groups if an ATE of 0 falls within the confidence interval for the difference. To quantify what the plots show, I computed the 95% CIs for the three pairs, and the results are shown below.

95% CI for ate_met_rosi: [-0.221197 -0.04058708]
95% CI for ate_met_lifestyle: [-0.14169061 0.04280872]
95% CI for ate_rosi_lifestyle: [-0.01043661 0.16943619]

As shown above, the only ATE that does not contain 0 is the 95% confidence interval for the average treatment effect of metformin–rosiglitazone. Therefore, the definitive conclusion from this trial is that the difference in success rate between metformin and rosiglitazone is statistically significant. In fact, because the 95% CI contains only negative values, we are 95% confident that the difference is negative, meaning the average success rate in those who received

rosiglitazone is roughly **4-22%** higher than the average success rate in those who took metformin.